Alice Gao

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Areas of interest

Human-computer interaction • UI Understanding • Computational Interfaces • Human-AI interaction • Data-driven design tools

Education

2023-*present* PhD in Computer Science & Engineering, **University of Washington**
2021-2023 MSc in Computer Science & Engineering, **University of Washington**
2017-2021 BS in Computer Science, **Princeton University**

Publications

* *denotes equal contribution*

Conference and Journal Papers

[c.2] **Alice Gao**, Andrew N. Meltzoff, Maarten Sap, Katharina Reinecke. Framing an AI with Values Reduces AI Reliance in AI-supported Writing Tasks. *ACM Conference on Fairness, Accountability, and Transparency (FAccT '26).* Montreal, QC, Canada. June 2026. [DOI]

[c.1] Nino Migineishvili, **Alice Gao**, Adinawa Adjabodjou, Dhanaraj Thaku, René Just, Katharina Reinecke. Regulating AI: Where U.S. State Policy and HCI (Mis)align. *ACM Conference on Human Factors in Computing Systems (CHI '26).* Barcelona, Spain. April 2026. [DOI]

[j.1] **Alice Gao**,* Samyukta Jayakumar,* Marcello Maniglia,* Brian Curless, Ira Kemelmacher-Shlizerman, Aaron R. Seitz and Steven M. Seitz. Don't Look at the Camera: Achieving Perceived Eye Contact. *Journal of Vision* 2025; 25(11):8. [DOI]

Workshop Papers & Posters

[w.1] K.J. Kevin Feng*, **Alice Gao***, Johanna Suvi Karrass*. Towards Semantically Aware Word Cloud Shape Generation. *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct).* Bend, OR. October 2022.

Preprints

[p.2] **Alice Gao**, Weixi Tong, Rishab Vempati, Katharina Reinecke, R. Benjamin Shapiro, Tianyi Zhang, Jason Wu. Training Computer Use Agents to Assess the Usability of Graphical User Interfaces.

April 2026. *In Submission.* [PDF]

[p.1] Donghoon Shin, **Alice Gao**, Rock Pang, Katharina Reinecke, Emily Tseng. Interrogating Design Homogenization in Web Vibe Coding. March 2026. *In Submission.* [PDF].

Research Experience

Jan 2024 - Present
Research Assistant, **University of Washington**
Advisor: R. Benjamin Shapiro
• Developed novel machine-learning algorithm to train computer use agents (CUAs) to automatically perform automated usability testing. [p.2]
• Developed comparison-based workflow using quantitative metrics for website designs to support UI/UX designers in creating website design for broader audiences.
• Designed a new interventions to reduce AI overreliance during AI-supported writing by up to 30%. [c.2]

Sep 2021- Jan 2024
Research Assistant, **GRAIL, University of Washington**
Advisors: Steve Seitz, Brian Curless, Ira Kemelmacher-Shlizerman
Developed machine learning techniques to create a model to redirect attention and eye gaze for a more attentive video conferencing experience. [j.1]

Sep 2019- Feb 2021
Undergraduate Research Assistant, **Princeton Vision & Learning Lab**, **Princeton University**
Advisor: Jia Deng; mentored by Hei Law
Improved associative embedding predictions and grouping from a neural network to provide faster and more accurate predictions for a one-stage object detector in instance segmentation.

Teaching

University of Washington (graduate)
Fall 2026 CSE442: Data Visualization, *Teaching Assistant*
Spring 2026 CSE446: Machine Learning, *Teaching Assistant*
Fall 2025 CSE373: Data Structures & Algorithms, *Teaching Assistant*
Spring 2025 CSE581: Computer Ethics, *Teaching Assistant*
Fall 2024 CSEP557: Current Trends In Computer Graphics, *Teaching Assistant*
Winter 2024 CSE581: Computer Ethics, *Teaching Assistant*

Princeton University (undergraduate)
Spring 2020 COS426: Computer Graphics, *Undergraduate Teaching Assistant*
Fall 2018-Fall 2019 COS226: Algorithms & Data Structures, *Course Grader*

Industry Experience

Jun-Aug 2020
Software Engineering Intern in Online Account Opening; **Capital One**; New York City, New York
Created new endpoint and unit tests, laid framework for streaming data platform, and launched new security groups.

Jun-Aug 2019 *AI Design Lab Intern*; **Tezign**; Shanghai, China
Assisted in optimizing image retrieval to match similar designs and contributed to training a model in judging strong vs weak graphic designs.

Jun-Aug 2018 *Technical and App Development Intern*; **Princeton Satellite Systems**; Princeton, New Jersey
Devised beginnings of gameplay for spacecraft simulation iOS app. Created 3D models for Phase II version of Direct Fusion Drive (DFD), a direct-drive, fusion-powered rocket engine.

Service

Reviewing: CHI 2024 – 2027, CHI LBW 2024, UIST 2026, FAccT 2026

Skills

Languages: Python, R, JavaScript, SQL, TypeScript, C, Go
Frameworks: PyTorch, scikit-learn, Pandas, React, Sveltekit, Node.js, Django/DRF, Flask
Tools: Docker, Apptainer, Figma
Other: Machine Learning, Quantitative and qualitative research methods, Statistical modeling, Experimental design, Semi-structured Interviews

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